Assignment 7D. Finite Element Method Marks 20

Posted on 29.10.2026 @ 2:30 pm and due on 29.10.2026 @ 5:40 pm

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1. Solve 1+1 dimension heat equation

$$\frac{\partial^2 u(x,t)}{\partial x^2} = \frac{\partial u(x,t)}{\partial t}$$

on a metal rod of length 2 units, with the initial conditions

$$\begin{aligned} u(0,t) = 0^o\text{C} = u(2,t) \qquad & \text{for} \quad 0 \le t \le 4 \\ u(x,0) = 20\,|\sin(\pi x)\,|^o\text{C} \qquad & \text{for} \quad 0 \le x \le 2 \end{aligned}$$

Use explicit scheme taking number of position grid $Nx = 20$ and time grid $Nt = 2000$. Show the temperature profile across the length of the rod at time steps 0, 10, 20, 50, 100, 200, 500 and 1000 in a plot. **[10]**

2. Solve the following PDE by Crank-Nicolson method. You have to work out analytically the matrix equation before you can implement it.

$$\begin{aligned} \frac{\partial u(x,t)}{\partial t} &= a\,\frac{\partial^2 u(x,t)}{\partial^2 x} + b\,\frac{\partial u(x,t)}{\partial x} + c\,u(x,t), \quad \text{with} \\ u(x,0) &= 2 + x^2, \;\; u(0,t) = 2 \;\text{ and }\; u(1,t) = 3 \end{aligned}$$

where $0 \le x \le 1$ and $a = 2,\ b = -1,\ c = 1$. You must upload the final algebraic equation and the corresponding matrix expression, like the one shown below (either hand-written or typed) along with your contour plots. **[10]**

The final matrix equation will look something like

$$V_j = \mathbb{A}^{-1}\,\mathbb{B}\,V_{j-1}, \quad \text{where}$$

$$\mathbb{A} = \begin{pmatrix} (1+2\alpha-\gamma) & (-\alpha-\beta) & 0 & \cdots & 0 \\ (-\alpha+\beta) & (1+2\alpha-\gamma) & (-\alpha-\beta) & \cdots & 0 \\ 0 & (-\alpha+\beta) & (1+2\alpha-\gamma) & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & 0 \\ \vdots & \vdots & \vdots & \ddots & (-\alpha-\beta) \\ 0 & 0 & \cdots & (-\alpha+\beta) & (1+2\alpha-\gamma) \end{pmatrix}$$

Matrix $\mathbb{B}$ will be somewhat similar.